Customer Shopping Behavior Analysis

1. Project Overview

This project explores real-world purchase activity recorded across 3,900 transactions, covering a wide range of product categories. The aim is to break down how different types of customers shop — who spends more, what drives their purchases, and which products consistently perform well.

Through 10 structured SQL queries, the analysis covers four main areas: spending differences across gender and age groups, customer segmentation based on purchase frequency, top-performing products within each category, and the role of subscriptions and discounts in shaping buyer behavior. Together, these findings help build a clearer picture of what influences purchasing decisions — and where businesses can focus their efforts to drive better outcomes.

2. Dataset Summary

- Rows: 3,900 - Columns: 18

- Key Features: - Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

● Data Loading: Imported the dataset using pandas.

● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

[7]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN

[8]:

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df.isnull().sum()
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Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
NaN	2	6	2	2	NaN	6	7
NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
NaN	2847	675	2223	2223	NaN	677	584
3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN

● Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

● Column Standardization: Renamed columns to snake case for better readability and documentation.

● Feature Engineering:

○ Created age_group column by binning customer ages.

○ Created purchase_frequency_days column from purchase data.

● Data Consistency Check: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79

Total rows: 839 Query complete 00:00

3.Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased text	average product rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Express	60.48
2	Standard	58.46

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customer bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	dicount_rate numeric
1	Shirt	0.00
2	Hat	0.00
3	Jacket	0.00
4	Sneakers	0.00
5	Gloves	0.00

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	number of customers bigint
1	returning	701
2	loyal	3116
3	new	83

8. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

	customer_id bigint	age bigint	gender text	item_purchased text	category text	purchase_amount bigint	location text	size text	color text	season text
1	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter
2	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter
3	3	50	Male	Jeans	Clothing	73	Massachuse...	S	Maroon	Spring
4	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring
5	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoi...	Spring
6	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer
7	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall
8	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter
9	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer
10	10	57	Male	Handbag	Accessori...	31	Missouri	M	Pink	Spring
11	11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall
12	12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter
13	13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter

Total rows: 3900 Query complete 00:00:00.243

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.

Key Insights & Recommendations

"Based on my analysis, I suggested:

- Promote subscriptions with exclusive benefits
- Introduce loyalty programs for repeat customers
- Optimize discount strategy to maintain profit margins
- Focus marketing on high-performing products
- Target high-revenue customer segments like specific age groups and express shipping users"